

# ECON 0100 | Demo C

Demos are similar to Checkpoints, often taken directly from past semesters. Work through the problems and check your work against mine. Practice answering clearly and completely. Show your work so someone else can understand your thought process. You are encouraged to work in small groups. Find a study room, grab some classmates, and work together on a whiteboard.

## Glittering Gum

Hogsmeade has recently seen a surge in shops selling Glittering Gum, a magical candy that makes the consumer's words sparkle and glisten as they speak. The Demand (marginal benefit) curve and Supply (marginal cost) curve for Glittering Gum can be represented using the following relationships:

$$D : P_b = 60 - Q_d \qquad S : P_s = 10 + Q_s$$

However, a magical externality has arisen: the widespread chewing of Glittering Gum has led to an increase in "Starlight Sneeze," a condition where non-consumers start sneezing out harmless but bothersome glitter. This condition imposes an inconvenience cost of 10 coins to both consumers and non-consumers in Hogsmeade.

## Q1 | Market Equilibrium (C3.1)

Use a graph and algebra to find equilibrium price, quantity, and DWL. Be sure to show your work.

a) Market Equilibrium Price: \_\_\_\_\_ Market Equilibrium Quantity: \_\_\_\_\_

b) Socially Efficient Quantity: \_\_\_\_\_ Deadweight Loss: \_\_\_\_\_

## Q2 | Policy Proposal (C4.1)

What type and size of policy would you propose to eliminate the DWL you identified in Question 1?

a) Policy Type: \_\_\_\_\_ Policy Size: \_\_\_\_\_

b) Buyer Price: \_\_\_\_\_ Seller Price: \_\_\_\_\_ Post-Policy Quantity: \_\_\_\_\_

## Q3 | Deadweight Loss Intuition (C3.1)

If the marginal cost of a gallon of gas is \$4.00, the negative externality is \$2.50, and the marginal benefit is \$4.50, what is the deadweight loss?

a) Deadweight Loss: \_\_\_\_\_